SoC Development Plan

Team : IIIT Bangalore

Introduction

The projects aims to design and implement a high-performance, 64-bit application-class processor capable of hosting a full Linux environment. By utilizing the open-source RISC-V ISA and integrating specialized arithmetic hardware, the project seeks to bridge the gap between general-purpose embedded computing and high-precision scientific workloads.

Main features:

- **64-bit RV64GC Core:** A robust 5-stage in-order pipeline implementing the standard "G" (General-purpose) and "C" (Compressed) instruction sets, providing the necessary complexity for modern operating systems while maintaining area efficiency.
- **1GHz Target Frequency:** Driven by an integrated on-chip PLL (Phase-Locked Loop).
- **Posit-Enabled ALU:** This allows the processor to handle higher dynamic range and precision compared to standard IEEE 754 floating-point units of the same bit-width.

Block Diagram

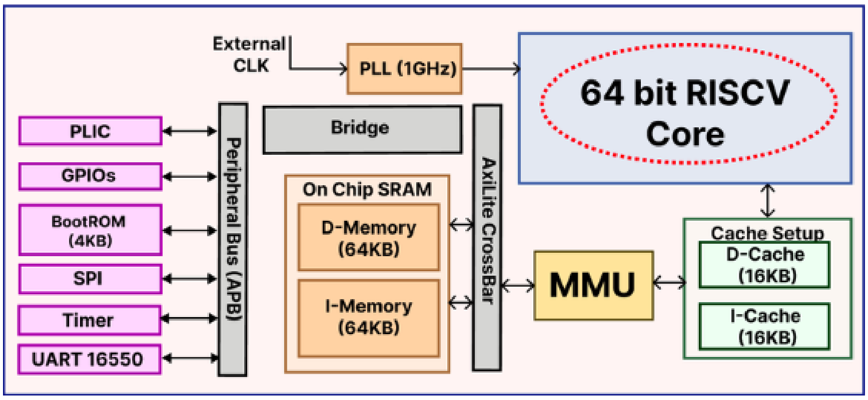
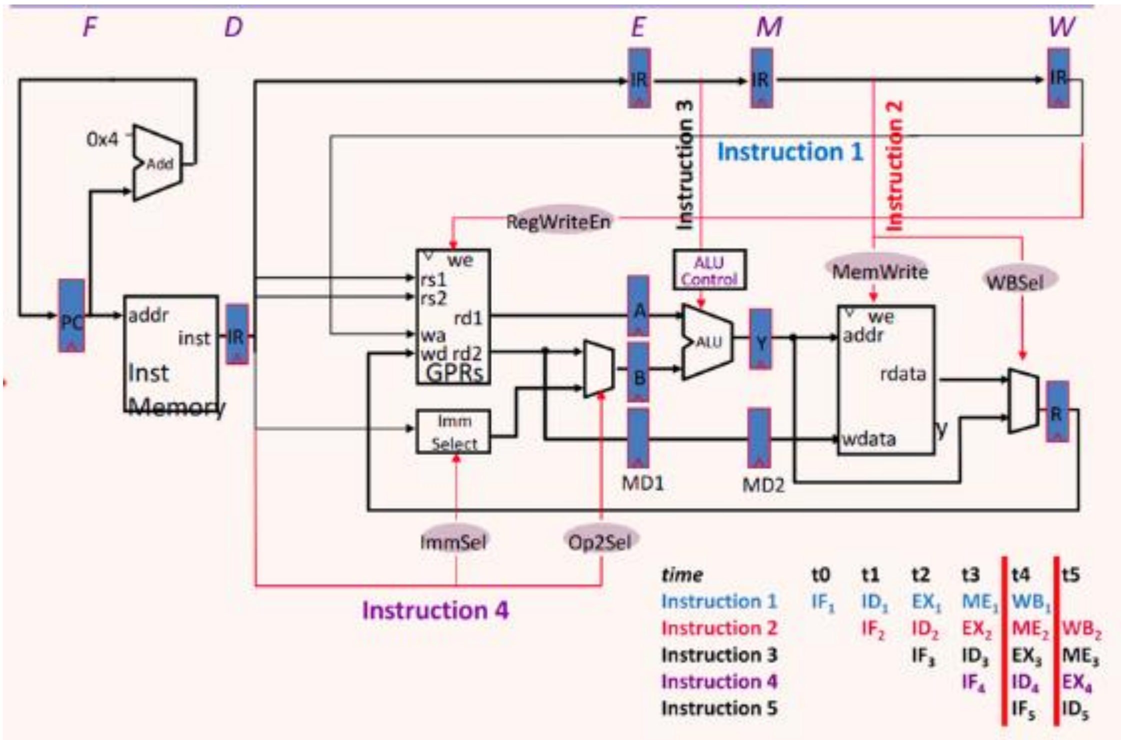


Figure 1: Simple top level illustration of the SoC



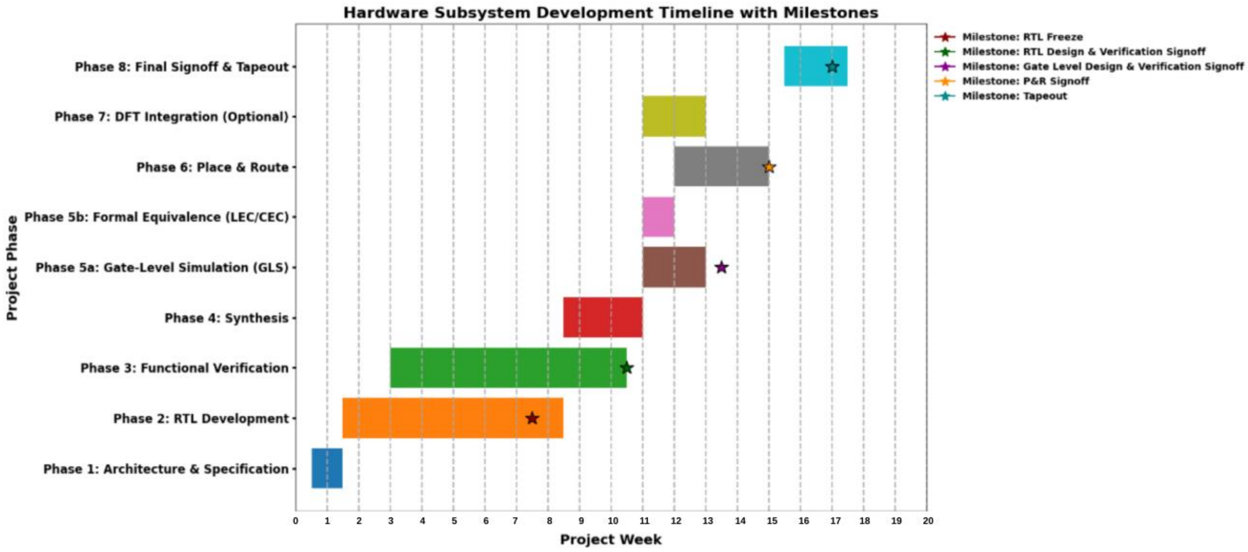
time	t0	t1	t2	t3	t4	t5
Instruction 1	IF_1	ID_1	EX_1	ME_1	WB_1	
Instruction 2		IF_2	ID_2	EX_2	ME_2	WB_2
Instruction 3			IF_3	ID_3	EX_3	ME_3
Instruction 4				IF_4	ID_4	EX_4
Instruction 5					IF_5	ID_5

Figure 2: 5 Stage pipeline showing the different stages.

List of IPs

1. **Sv39/48 MMU :** Hardware logic for virtual-to-physical address translation.
2. **PLIC (Platform-Level Interrupt Controller):** Manages external interrupts from all peripherals.
3. **Instruction SRAM (64KB):** Dedicated on-chip memory for code.
4. **Data SRAM (64KB):** Dedicated on-chip memory for data storage.
5. **BootROM (4KB) with SPI :** Read-only memory for the primary bootloader.
6. **16KB I-Cache and 16KB D-Cache:** Direct Mapped, to reduce core stalls.
7. **PLL :** On-chip Phase-Locked Loop for 1GHz clock synthesis.
8. **UART 16550 :** Standard serial communication for console/debugging.
9. **Communication:** Host- AxiLite ; Peripherals – APB
10. **Timer and GPIOs**

Timelines



Phase 1: Architecture & Specification

Tasks & Checks:

- Define requirements (CPU integration, peripherals, memory, bus, debug).
- Clock/reset/power domain plan.
- Verification plan – Functional, Coverage, GLS
- Tools and Simulation planning

Deliverables: Arch spec, block diagram, memory map, domain diagrams, initial verification plan.

Phase 2: RTL Development

Tasks & Checks:

- RTL coding of CPU subsystem, interconnect, memory, peripherals.
- Early CDC/RDC static checks.
- Synthesis sanity checks (compile-only).
- Unit-level directed tests.

Weekly Deliverables: Lint-clean, synthesis-ready RTL, passing block-level unit tests.

Phase 3: Functional Verification

Tasks & Checks:

- UVM testbench, constrained-random + directed tests.
- Assertions (SVA for protocols, resets).
- Coverage closure (>95% functional, >90% code).
- Formal checks on FIFOs, arbiters, deadlock freedom.
- UPF/power-aware simulation.
- Continuous regression + nightly coverage tracking.

Deliverables: Weekly Verification report, Coverage closure.

Phase 4: Synthesis

Tasks & Checks:

- Synthesis with constraints (multi-corner/mode).
- STA (setup/hold).
- Area & power vs budget.
- CDC/RDC re-check at netlist level.
- LEC: RTL ↔ Synth netlist.
- Functional Simulation Checks

Deliverables: Timing-clean synthesized netlist, LEC report, synthesis reports.

Phase 5a: Gate-Level Simulation

Tasks & Checks:

- Zero-delay GLS.
- SDF GLS (post-route timing).
- X-Prop GLS (reset/powerup).
- Scan/MBIST GLS sims.
- UPF-aware GLS (power gating sequences).

Deliverables: GLS regression report, clean reset/power sequences

Phase 5b: Formal Equivalence (LEC/CEC)

Tasks & Checks:

- RTL ↔ Synth netlist.
- Synth ↔ Scan netlist.
- Scan ↔ P&R netlist.
- ECO LEC re-checks.

Deliverables: LEC reports (0 mismatches).

Phase 6: Place & Route

Tasks & Checks:

- Floor Planning, power grid.
- Placement, CTS, routing.
- STA across corners.
- IR-drop/EM analysis.
- Congestion checks.
- ECO iterations for timing/power closure.

Deliverables: Timing-closed post-route netlist + GDS.

Phase 7: DFT Integration (Optional)

Tasks & Checks:

- Scan insertion.
- MBIST insertion.
- ATPG pattern generation.
- Coverage closure (>95% stuck-at, >90% transition).
- GLS with scan + BIST.

Deliverables: DFT reports, scan/MBIST verified netlist.

Phase 8: Final Signoff & Tapeout

Checks:

- STA signoff (all corners, all modes).
- DRC/LVS clean (Calibre).
- IR-drop/EM signoff.
- Functional coverage final closure.
- DFT coverage signoff.
- Xprop signoff (RTL + GLS).
- GLS regressions fully passing.
- CDC/RDC clean.
- Tapeout package (GDSII, reports, documentation).

Deliverables: Tapeout release package to foundry.